Code HW #6:

Submit your solution on Canvas (see end for submission checklist).

For the following questions, consider the Stanford Manipulator shown in Figure 1.

The local frames of each link are shown as $x_0 y_0 z_0$ for the base link to $x_6 y_6 z_6$ for end-effector link 6. The joint axes are about the z axis of the previous link, such that $\boldsymbol{\omega}_1 = z_0$, $\boldsymbol{\omega}_2 = z_1$, etc. Each axis is a revolute joint, except for $\boldsymbol{\omega}_3$, which is prismatic.

For the distances d_{1-6} use the following:

$d_1 = 40$ cm
$d_2 = 50$ cm
$d_3 = 60$ cm
$d_6 = 30$ cm

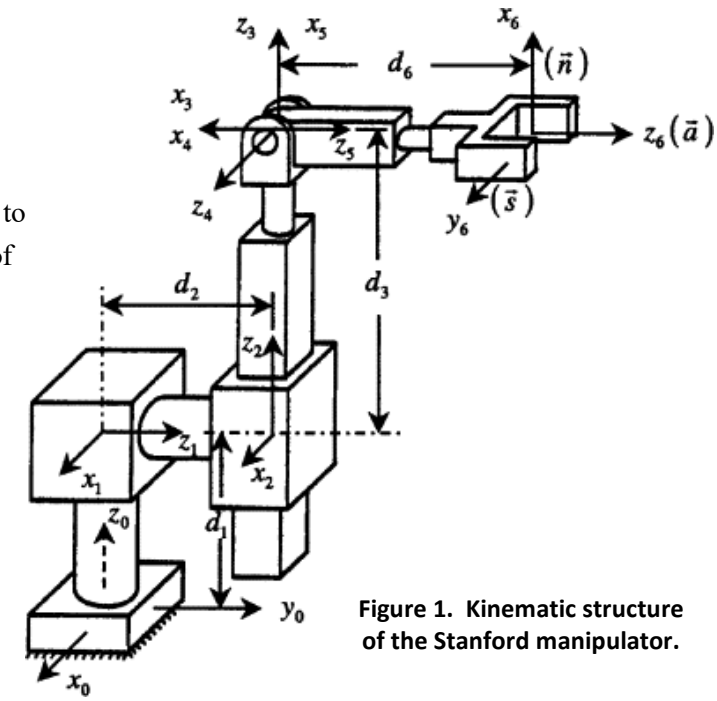

Figure 1. Kinematic structure of the Stanford manipulator.

1) Using MATLAB or Python, find the spatial manipulator Jacobian, $\mathbf{J}_{st}(\theta)$, of the Stanford manipulator above. Use the posted sub-functions. Use the simplify command. Your script should only output the symbolic expression `JstthCKB`. Use the `matlabFunction` command to create a function called `JstthCKB.m` (use your initials). This function should have a single vector of all 6 angles as the input.
2) In the same script used for the spatial manipulator Jacobian, use the `matlabFunction` or `lambdify` commands to create numeric functions for the forward kinematics for each link: $\mathbf{g}_{s1}(\theta)$, $\mathbf{g}_{s2}(\theta)$, $\mathbf{g}_{s3}(\theta)$, $\mathbf{g}_{s4}(\theta)$, $\mathbf{g}_{s5}(\theta)$, & $\mathbf{g}_{st}(\theta)$. Each function should have a single vector of all 6 angles as the input.
3) In a second script using these functions and the sample code posted on Canvas, animate the robot going through a motion described by the equations below. You may choose the width of each link, etc., but keep the dimensions given above. Although some joint axes are coincident, create a rectangular prism for each link so that the relative motions can be seen clearly.

$$\theta_1 = \frac{3\pi}{4} + \frac{\pi}{4}\sin\left(\frac{\pi}{2}t\right) \qquad \theta_2 = \frac{\pi}{2}\sin\left(\frac{\pi}{2}t\right) \qquad \theta_3 = \frac{d_3}{2}\sin(\pi t)$$

$$\theta_4 = \frac{\pi}{2}\sin\left(\frac{\pi}{2}t\right) \qquad \theta_5 = \frac{\pi}{2} + \frac{\pi}{2}\sin(\pi t) \qquad \theta_6 = \frac{\pi}{2}\sin\left(\frac{\pi}{2}t\right)$$

4) Once working correctly, record a video of the motion for 0 to 4 seconds using `VideoWriter`.
5) In the same animation file, find the position and velocity of the origin of the tool frame. Create a single figure with the following plots:
   a. TOP PLOT: x, y, and z position of the origin of frame 6 vs time for 0 to 4 seconds. Use a red line for the x-position, a green line for the y-position, and a blue line for the z-position. Make sure to include a legend and label the axes.
   b. BOTTOM PLOT: x, y, and z velocity of the origin of frame 6 vs time for 0 to 4 seconds. Use a red line for the x-velocity, a green line for the y-velocity, and a blue line for the z-velocity. Make sure to include a legend and label the axes.

HW #6: WHAT TO SUBMIT

Submit ***individual files*** on Canvas (not zipped) using the naming convention given.

A. **HW06_Lastname_Kinematics.m:** Your kinematics m-file. This script should create 7 sub-functions as defined above.
B. **HW06_Lastname_Animation.m:** Your animation m-file.
C. **HW06_Lastname_Video**. Your video of the animation.
D. **HW06_Lastname_posvelplot.pdf**. The full-page figure with a top and bottom plot as described above.